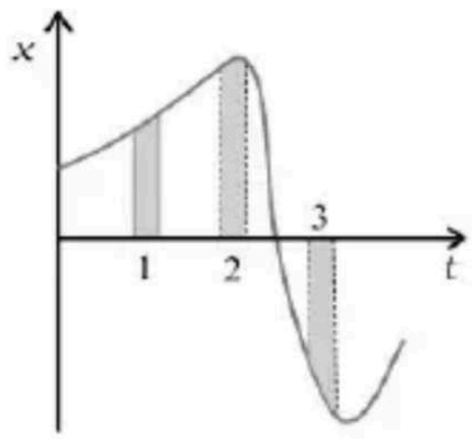
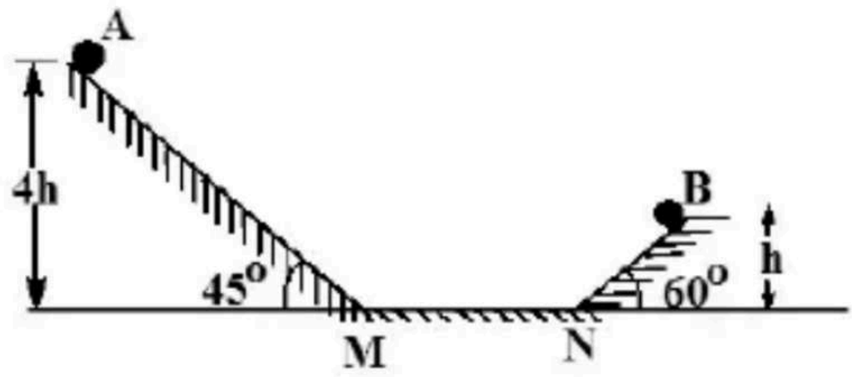
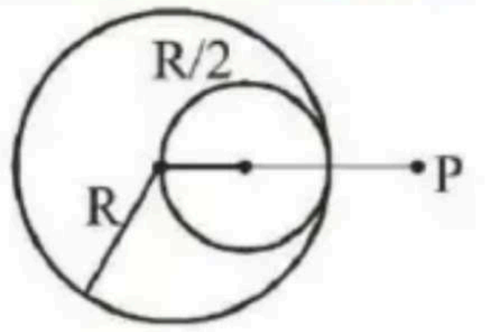
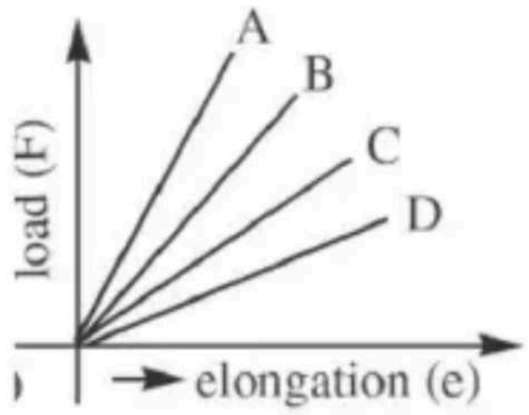
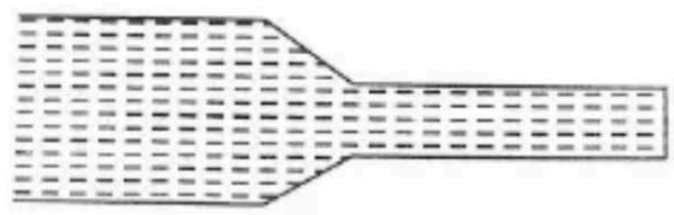
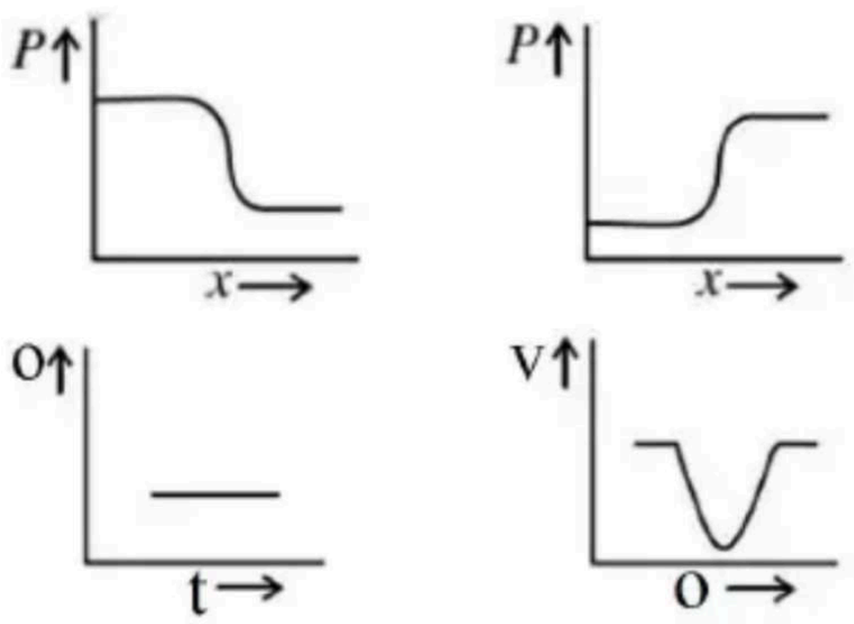

x
1
2
3
t

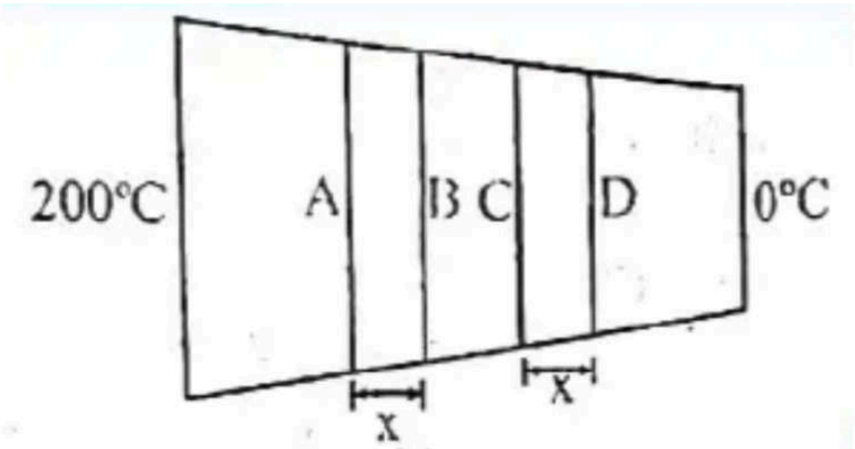

A
4h
45°
M
N
B
h
60°

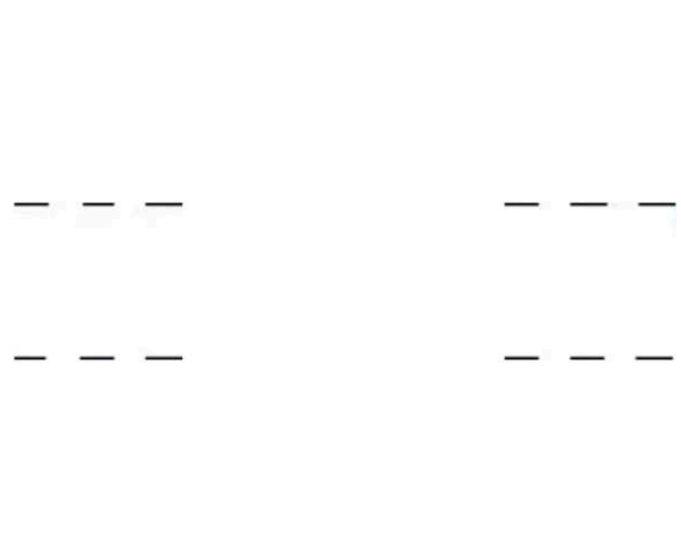

R/2
R
P

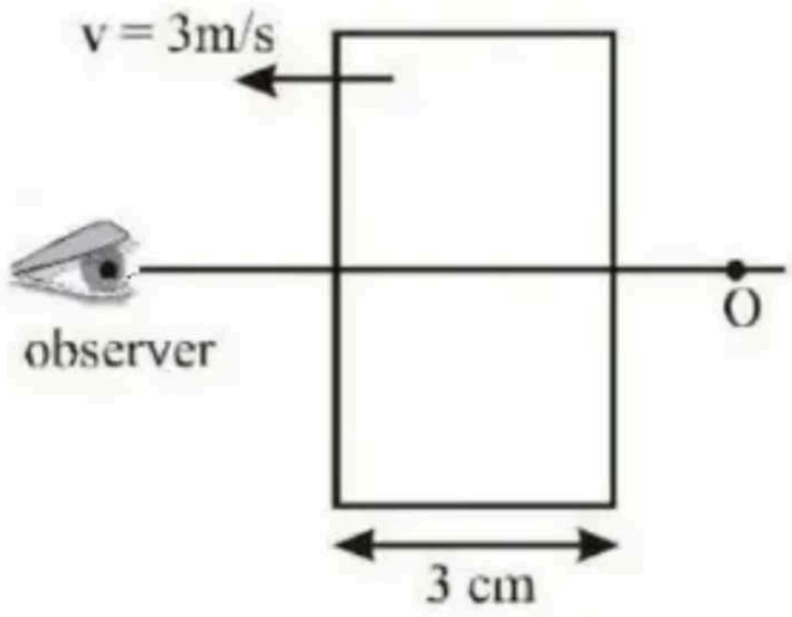

A
B
C
D
load (F)
elongation (e)



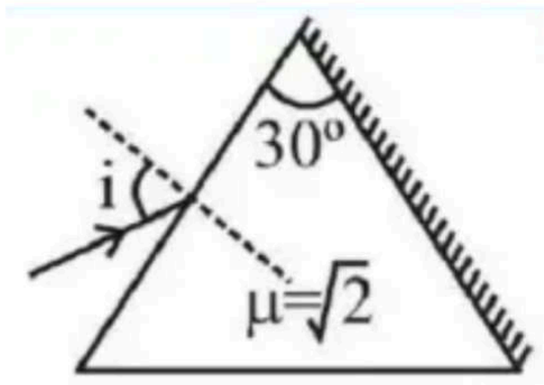

P
x
P
x
O
t
V
O


200°C
A
B C
D
0°C
x
x


C
B
2
V(m³)
1
A
P(N/m²)
10


v = 3m/s
observer
O
3 cm

3D 4D



30° 60°

45°

—

—

—

—

4500Å°

30°

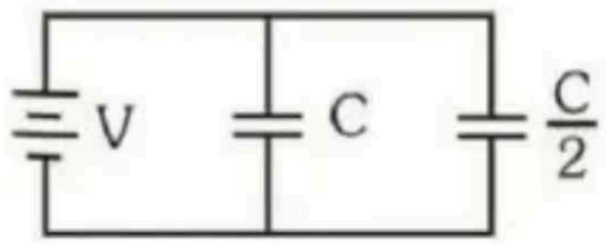
$\sin^{-1}\left(\frac{3}{5}\right)$ $\sin^{-1}\left(\frac{3}{4}\right)$

$\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ $\sin^{-1}\left(\frac{4}{3}\right)$

10μC

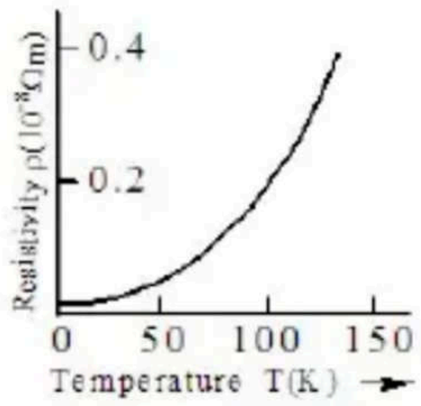
30°

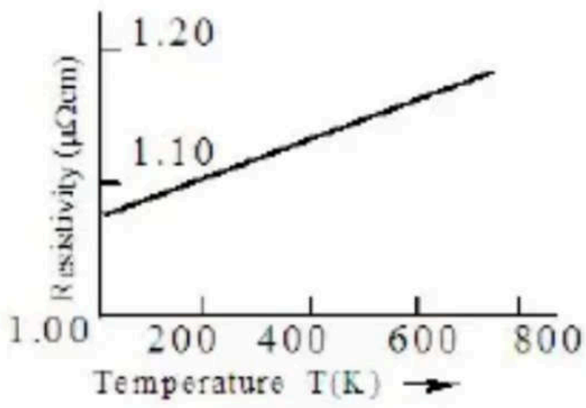
$\frac{C}{2}$

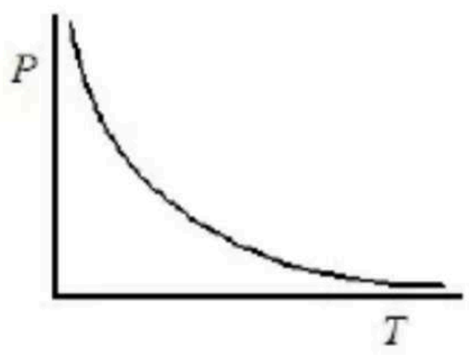


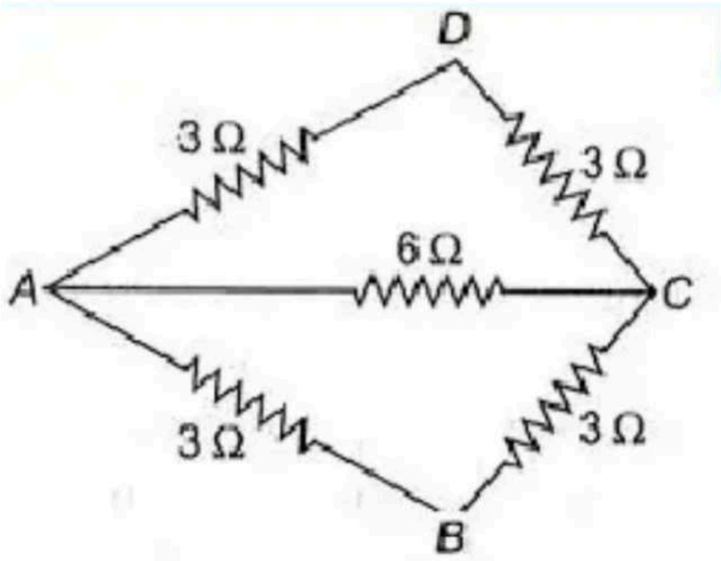
$\frac{1}{2}CV^2$ $2CV^2$

$\frac{3}{4}CV^2$ $\frac{3}{2}CV^2$









5 Ω

3 Ω

Ω

15 10^{5} J / T

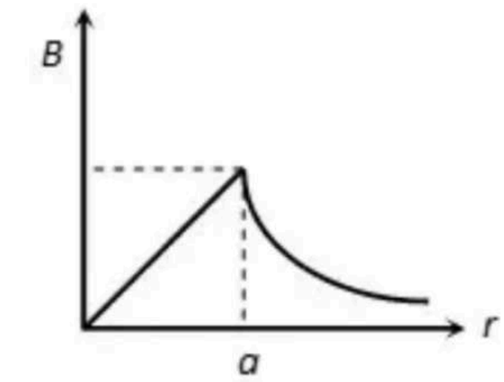
80×10^{3} A / m

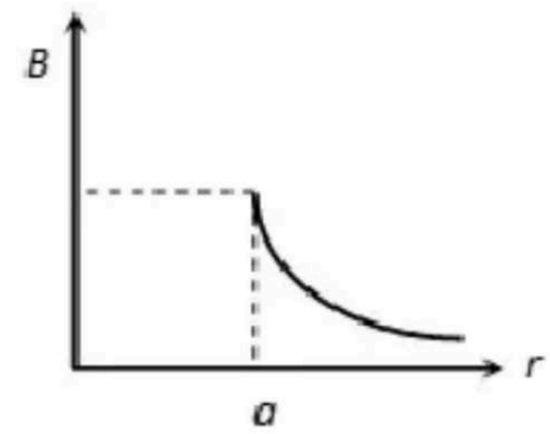
1.8×10^{-4} 2.7×10^{-4}

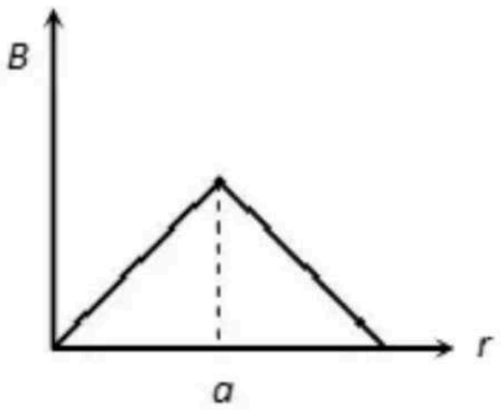
1.8×10^{-3} 2.7×10^{-3}

$\frac{B}{81}$

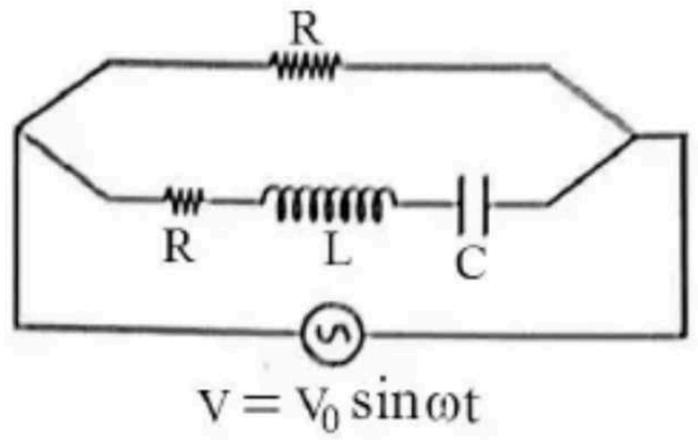
$\frac{B}{243}$











$V = V_0 \sin\omega t$

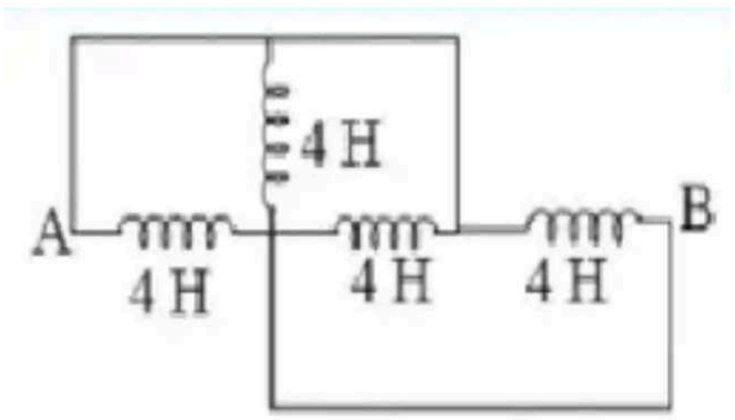
$\frac{V_0\sqrt{2}}{2R}$

$\sqrt{}$ $\frac{4V_0\sqrt{2}}{R}$

37. (A) All isotopes of elements are not stable.
(B) Orbital angular momentum of an electron of hydrogen atom is independent of the radius of orbit.
(C) In hydrogen atom, if the zero potential energy of electron is choosen differently then kinetic energy of the electron in its orbit does not change.
(1) A, B are correct (2) B, C are correct
(3) A, C are correct (4) Only A is correct

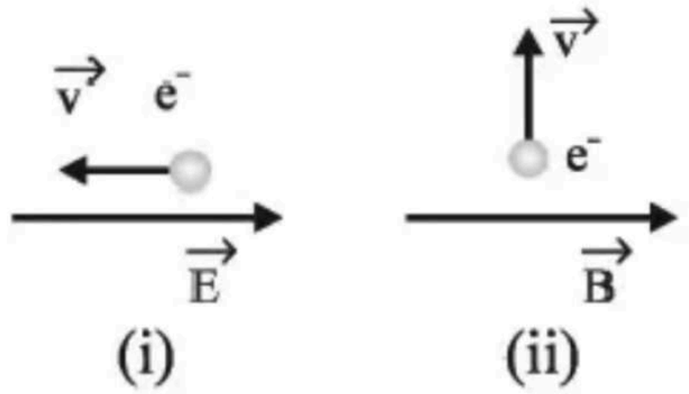
38. A graph is drawn between stopping potential as a function of frequency of incident light, for a given surface. The intercept on stopping potential axis will be (all the terms with usual meaning)
(1) h (2) h/e
(3) W (4) W/e

39. The equivalent inductance between A and B is



(1) 0.8 H (2) 1 H
(3) 4 H (4) 16 H

40. An electron is moving through a field. It is moving
(i) opposite to an electric field
(ii) perpendicular to a magnetic field as shown. For each situation the de-Broglie wave length of electron



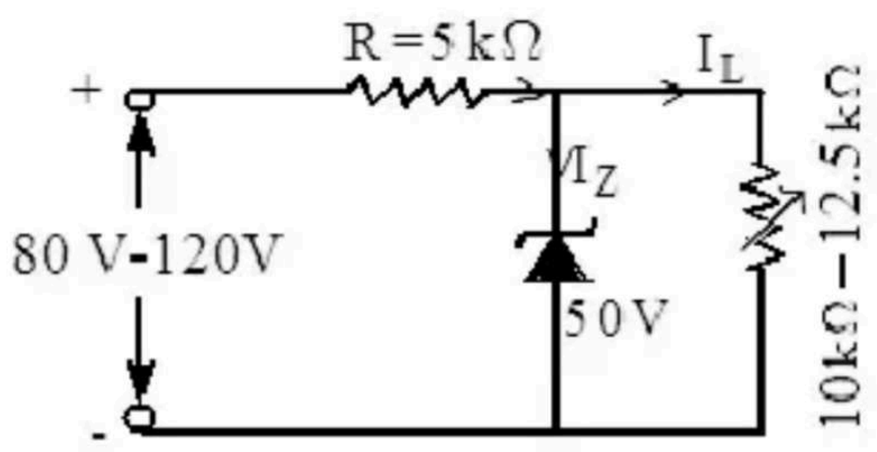
(1) Increasing, increasing
(2) Increasing, decreasing
(3) Decreasing, same
(4) Increasing, Same

41. A photon and an electron have equal energy E. Then $\left(\lambda_{photon} / \lambda_{electron}\right)$ is proportional to
(1) $\sqrt{E}$
(2) $1/\sqrt{E}$
(3) $1/E$
(4) Does not depend upon E

42. Given below are two statements Assertion (A) and Reason (R).
Assertion-(A): The density of all the nuclei is almost same.
Reason-(R):Radius of nucleus is inversly proportional to the cube root of mass number of the nucleus .
In the light of the above statements, choose the correct answer from the options given below.
(1) Both 'A' and 'R' are true and 'R' is the correct explanation of 'A'.
(2) Both 'A' and 'R' are true and 'R' is not the correct explanation of 'A'
(3) 'A' is true and 'R' is false
(4) A is false, but R is true

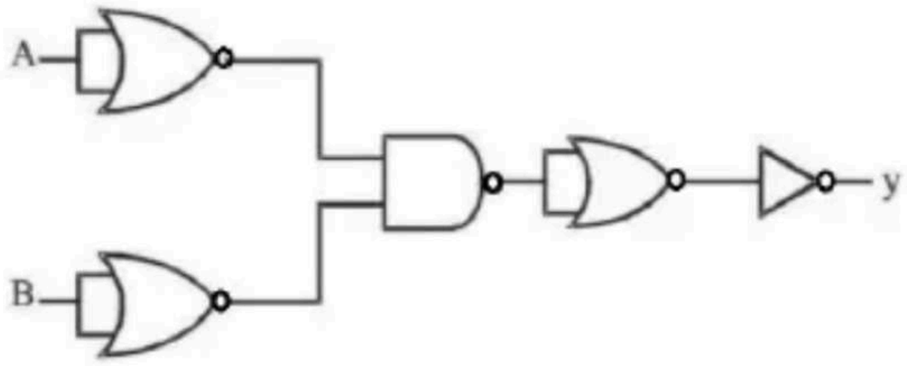
43. In an apparatus the electric field was found to oscillate with an amplitude of $18 Vm^{-1}$. The rms value of the oscillating magnetic field is
(1) 6×10^{-8} T (2) 4.23×10^{-8} T
(3) 9×10^{-8} T (4) 8.4×10^{-8} T

44. Find the maximum zener current in the given voltage regulator



(1) 9 mA (2) 12 mA
(3) 2 mA (4) 10 mA

45. Which logic gate is represented by the following combination of logic gates



(1) AND (2) NAND
(3) NOR (4) OR

CHEMISTRY

46. The velocity of particle A is $0.1 ms^{-1}$ and that of particle B is $0.05\ ms^{-1}$. If the mass of particle B is five times that of particle A, then the ratio of de Broglie wavelengths associated with the particles A and B is.
(1) 2 : 5 (2) 3 : 4
(3) 6 : 4 (4) 5 : 2

47. Given below are the quantum numbers for four electrons.
(A) $n = 3, l = 2, m_l = 1, m_s = +1/2$
(B) $n = 4, l = 1, m_l = 0, m_s = +1/2$
(C) $n = 4, l = 2, m_l = -2, m_s = -1/2$
(D) $n = 3, l = 1, m_l = -1, m_s = +1/2$
The correct order of increasing energy is:
(1) D < B < A < C (2) D < A < B < C
(3) B < D < A < C (4) B < D < C < A

48. Identify the incorrect match.

NAME		IUPAC Official Name	
A	Unnilunium	i	Mendelevium
B	Unniltrium	ii	Lawrencium
C	Unnilhexium	iii	Seaborgium
D	Ununbium	iv	Darmstadtium

(1) (B), (ii) (2) (C), (iii)
(3) (D), (iv) (4) (A), (i)

49. Which of the given elements A, B, C, D and E with atomic number 2, 3, 7, 10 and 30 respectively belong to the same period?
(1) A, B, C (2) B, C, D
(3) A, D, E (4) B, D, E

50. Which one of the following is not correct about hybridisation of orbitals?
(1) The orbitals present in the valence shell of the atom get hybridized.
(2) The orbitals undergoing hybridization have almost equal energy.
(3) The hybrid orbitals formed have same shape and energy.
(4) Pure atomic orbitals are more effective in forming stable bonds than hybrid orbitals.

51. The increasing order of bond order among the following is
$O_2^+, O_2, O_2^-, O_2^{2-}$.
(1) $O_2^{2-} < O_2^- < O_2 < O_2^+$
(2) $O_2^- < O_2^{2-} < O_2 < O_2^+$
(3) $O_2^- < O_2^{2-} < O_2^+ < O_2$
(4) $O_2^- < O_2^+ < O_2^{2-} < O_2$

52. Assertion: In NH_3, N is sp^3 hybridised, but bond angle is found to be 107°.
Reason: The decrease in bond angle in NH_3 is due to repulsion between the lone pair – bond pair of central atom.
(1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
(2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
(3) Assertion is correct but Reason is incorrect.
(4) Assertion is incorrect and Reason is correct.

53. In a compound C, H and N atoms are present in 9:1:3.5 by weight. Molecular weight of compound is 108. Molecular formula of compound is
(1) $C_2H_6N_2$ (2) C_2H_4N
(3) $C_6H_8N_2$ (4) $C_9H_{12}N_3$

54. 20.0 kg of $N_2(g)$ and 3.0 kg of $H_2(g)$ are mixed to produce $NH_3(g)$. The amount of $NH_3(g)$ formed is
(1) 17 kg (2) 8.4 kg
(3) 20 kg (4) 3 kg

55. The bond dissociation energies of X_2, Y_2 and XY are in the ratio of 1:0.5:1. ΔH for the formation of XY is $-200\ kJ\ mol^{-1}$. The bond dissociation energy of X_2 will be
(1) $200\ kJ\ mol^{-1}$ (2) $100\ kJ\ mol^{-1}$
(3) $600\ kJ\ mol^{-1}$ (4) $800\ kJ\ mol^{-1}$

56. Match Column-I with Column-II.

Column-I		Column-II	
A	ΔH = -ve; ΔS = -ve ΔG = -ve	p	Reaction will be non-spontaneous at high temperature
B	ΔH = -ve; ΔS = -ve ΔG = +ve	q	Reaction will be non-spontaneous at low temperature
C	ΔH = +ve; ΔS = +ve ΔG = +ve	r	Reaction will be spontaneous at low temperature
D	ΔH = +ve; ΔS = +ve ΔG = -ve	s	Reaction will be spontaneous at high temperature

(1) A – (q), B – (r), C – (p), D – (s)
(2) A – (r), B – (p), C – (q), D – (s)
(3) A – (r), B – (q), C – (s), D – (p)
(4) A – (q), B – (s), C – (p), D – (r)

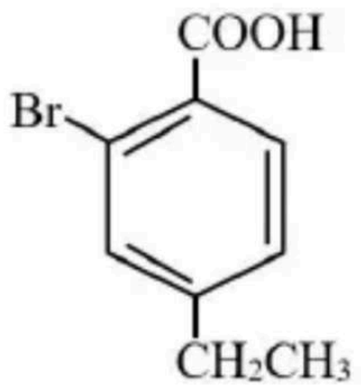
57. Statement-I: The value of equilibrium constant is independent of initial concentrations of the reactants and products.
Statement-II: Equilibrium constant is temperature dependent.
(1) Both statement I and II are correct.
(2) Both statement I and II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement II is correct but statement I is incorrect.

58. Arrange the following solutions in the decreasing order of pH:
(A) 0.01 M HCl
(B) 0.01 M NaOH
(C) 0.01 M CH_3 COONa
(D) 0.01 M NaCl
(1) (A) > (C) > (D) > (B)
(2) (A) > (D) > (C) > (B)
(3) (B) > (C) > (D) > (A)
(4) (B) > (D) > (C) > (A)

59. pH of a saturated solution of $Ba(OH)_2$ is 12. The value of solubility product (K_{sp}) of $Ba(OH)_2$ is:
(1) 3.3×10^{-7} (2) 5.0×10^{-7}
(3) 2.0×10^{-4} (4) 5.0×10^{-9}

60. Which of the following sequence of T and F is correct for given statements? Here T stands for true and F stands for false statement.
(i) The tendency to show +2 oxidation state increase in the sequence Ge < Sn < Pb.
(ii) Tin in +2 state is a reducing agent.
(iii) Lead compounds in +2 state are strong oxidising agents.
(iv) In tetravalent state, molecules of group 14 elements act as electron donor species.
(1) TTTT (2) TTFF
(3) TTFT (4) TFFT

61. What is the IUPAC name of the following compound?



(1) 6–Bromo–4–ethylbenzene carboxylic acid
(2) 2–Bromo–4–ethylbenzene carboxylic acid
(3) Ortho–bromo–paraethyl benzoic acid
(4) 4–Bromo–3–ethyl benzoic acid

62. Match Column-I with Column-II.

Column-I		Column-II	
A	CH_3COOH & $HCOOCH_3$	p	Functional isomers
B	1 butene & 2-butene	q	Metamers
C	diethyl ether & methyl propyl ether	r	Position isomers
D	Butane, 2-methyl propane	s	Chain isomers

(1) A – (p), B – (r), C – (q), D – (s)
(2) A – (q), B – (r), C – (s), D – (p)
(3) A – (q), B – (s), C – (p), D – (r)
(4) A – (q), B – (p), C – (s), D – (r)

63. Assertion: Simple distillation can help in separating a mixture of chloroform (boiling point 61°C) and aniline (boiling point 184°C).

Reason: Liquids with a difference of more than 40°C in their boiling points can be separated by simple distillation.

(1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect and Reason is correct.

64. In the given reactions X, Y, Z are:

(i) $C_2H_5I + 2H \xrightarrow[C_2H_5OH]{Zn-Cu} X$

(ii) $CH_3COONa + H_2O \xrightarrow{Electrolysis} Y$

(iii) $CH_3CH_2CH_2CH_2CH_2CH_3 \xrightarrow[773K,10-20atm]{Cr_2O_3/Al_2O_3} Z$

(1) $X = C_2H_6, Y = C_2H_6, Z = C_6H_6$

(2) $X = CH_4, Y = CH_3COOH, Z = CH_3CH_3$

(3) $X = C_2H_6, Y = CH_4, Z = C_4H_{10}$

(4) $X = C_2H_6, Y = CH_4, Z = C_5H_{10}$

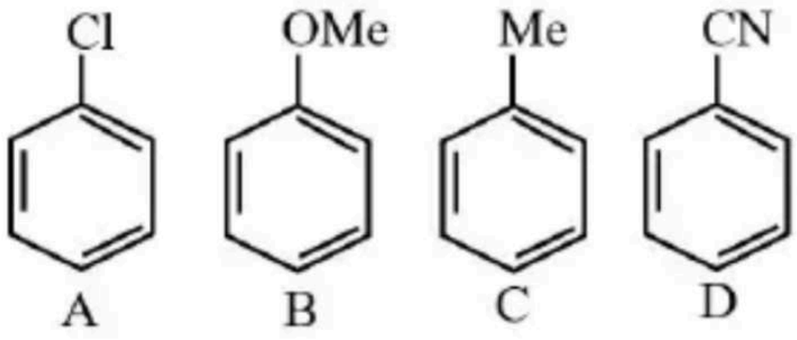
65. Choose the correct regents used in the conversion.

$$CH_2 = CH_2 \xrightarrow{(p)} \underset{\substack{| \\ Br}}{CH_2} - \underset{\substack{| \\ Br}}{CH_2} \xrightarrow{(q)}$$

$$H_2C = CHBr \xrightarrow{(r)} HC \equiv CH \xrightarrow{(s)}$$ 

	(p)	(q)	(r)	(s)
(1)	Br_2	alc.KOH	NaOH	Al_2O_3
(2)	HBr	alc.KOH	CaC_2	$KMnO_4$
(3)	HBr	alc.KOH	$NaNH_2$	red hot iron tube
(4)	Br_2	alc.KOH	$NaNH_2$	red hot iron tube

66. The increasing order of reactivity of the following compounds towards aromatic electrophilic substitution reaction is:



(1) D < A < C < B (3) B < C < A < D

(3) A < B < C < D (4) D < B < A < C

67. The correct statements among the following are:

(i) The vapour pressure of a liquid decreases with increase of temperature.

(ii) The liquid boils at the temperature at which its vapour pressure is equal to the atmospheric pressure.

(iii) Vapour pressure of the solvent decreases by the addition of non-volatile solute.

(iv) Vapour pressure of the pure solvent and solution is a function of temperature.

(1) (i), (ii) and (iv) are correct

(2) (i), (iii) and (iv) are correct

(3) (ii), (iii) and (iv) are correct

(4) (i), (ii) and (iii) are correct

68. Assertion: When a solution is separated from the pure solvent by a semi-permeable membrane, the solvent molecules pass through it from pure solvent side to the solution side.

Reason: The flow of solvent occurs from a region of high concentration of solution to a region of low concentration of solution.

(1) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(2) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect and Reason is correct.

69. The reduction potential of half-cell consisting of zinc electrode in 0.01 M $ZnSO_4$ solution at 25°C is $\left(E^{o}_{RP} = -0.763\ V\right)$

(1) –0.8221 V (2) –8.221 V
(3) –0.5282 V (4) –9.282 V

70. Which of the following pair(s) is/are incorrectly matched?
(i) R (resistance) – ohm
(ii) ρ (resistivity) – ohm metre
(iii) G (conductance) – ohm metre^{-1}
(iv) κ (conductivity) - seimens metre^{-1}
(1) (i), (ii) and (iii) (2) (ii) and (iii)
(3) (i), (ii) and (iv) (4) (iii) only

71. The number of coulombs required to reduce 12.3 g of nitrobenzene to aniline is:
(1) 115800 C (2) 5790 C
(3) 28950 C (4) 57900 C

72. During the kinetic study of the reaction, 2A + B → C + D, following results were obtained:

Run	[A] (mol L^{-1})	[B] (mol L^{-1})	Initial rate of formation of D (mol $L^{-1}min^{-1}$)
I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

Based on the above data which one of the following is correct?
(1) rate =k $[A]^2[B]$ (2) rate =k [A] [B]
(3) rate =k $[A]^2[B]^2$ (4) rate =k [A] $[B]^2$

73. Statement-I: In Arrhenius equation, $e^{-E_a/RT}$ corresponds to the fraction of molecules that have kinetic energy greater than E_a.
Statement-II: E_a can be calculated as follows

$$\log\frac{k_1}{k_2} = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1T_2}\right].$$

(1) Both statement I and II are correct.
(2) Both statement I and II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement II is correct but statement I is incorrect.

74. Statement-I: BiH_3 is the strongest reducing agent amongst the hydrides of nitrogen family.
Statement-II: P_2O_3 is more acidic than P_2O_5.
(1) Both statement I and II are correct.
(2) Both statement I and II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement II is correct but statement I is incorrect.

75. Which one of the following arrangements does not give the correct picture of the trends indicated against it?
(i) $F_2 > Cl_2 > Br_2 > I_2$: Oxidizing power
(ii) F > Cl > Br > I: Negative electron gain enthalpy
(iii) $F_2 > Cl_2 > Br_2 > I_2$: Bond dissociation energy
(iv) F > Cl > Br > I: Electronegativity.
(1) (ii) and (iv) (2) (i) and (iii)
(3) (ii) and (iii) (4) (ii), (iii) and (iv)

76. Gadolinium belongs to 4f series. Its atomic number is 64. Which of the following is the correct electronic configuration of gadolinium?
(1) $[Xe]4f^7\ 5d^1\ 6s^2$ (2) $[Xe]4f^6\ 5d^2\ 6s^2$
(3) $[Xe]4f^8\ 6d^2$ (4) $[Xe]4f^9\ 5s^1$

77. Identify the incorrect statement.
(1) The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.
(2) Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.
(3) The oxidation states of chromium in CrO_4^{2-} and $Cr_2O_7^{2-}$ are not the same.
(4) $Cr^{2+}\left(d^4\right)$ is a stronger reducing agent than $Fe^{2+}\left(d^6\right)$ in water.

78. The correct order of energy of absorption for the following metal complexes is

A: $\left[Ni(en)_3\right]^{2+}$ B: $\left[Ni(NH_3)_6\right]^{2+}$

C: $\left[Ni(H_2O)_6\right]^{2+}$

(1) C < B < A (2) B < C < A

(3) A < B < C (4) A < C < B

79. The compound used in the treatment of lead poisoning is:

(1) D-penicillamine (2) desferrioxime B

(3) C*is*-platin (4) EDTA

80. The most reactive compound towards S_N1 reaction is:

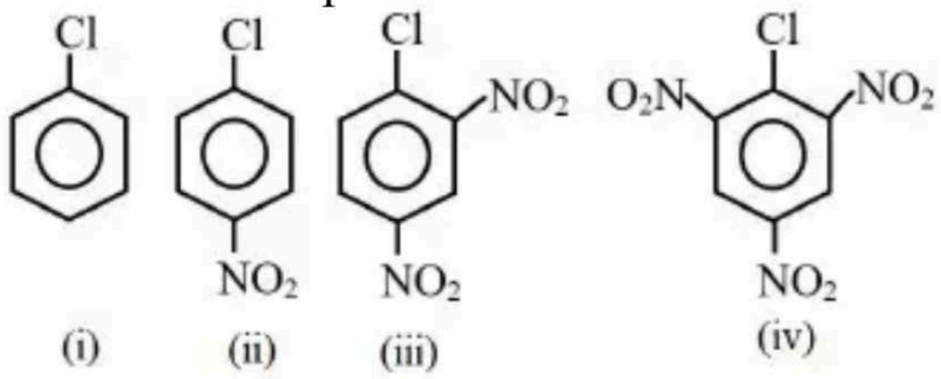
(1) $C_6H_5CH(C_6H_5)Br$

(2) $C_6H_5CH(CH_3)Br$

(3) $C_6H_5C(CH_3)(C_6H_5)Br$

(4) $C_6H_5CH_2Br$

81. The correct order of the following compounds showing increasing tendency towards nucleophilic substitution reaction is:



(1) (i) < (iv) < (iii) < (ii)

(2) (iv) < (i) < (ii) < (iii)

(3) (iv) < (i) < (iii) < (ii)

(4) (i) < (ii) < (iii) < (iv)

82. In the following sequence of reactions,

$$CH_3CH_2OH \xrightarrow{P+I_2} A \xrightarrow[ether]{Mg} B \xrightarrow{HCHO} C \xrightarrow{H_2O} D$$

the compound is

(1) Propanal (2) Butanal

(3) n-butyl alcohol (4) n-propyl alcohol

83. Consider the following reaction:

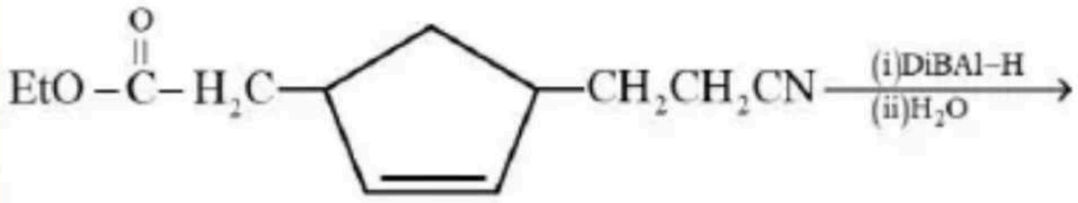
$$Phenol \xrightarrow{Zn\ dust} X \xrightarrow[Anhydrous\ AlCl_3]{CH_3Cl} Y \xrightarrow[H_3O^+]{Alk.\ KMnO_4,\Delta} Z.$$

The product Z is

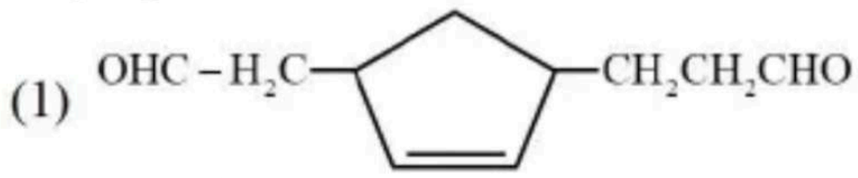
(1) Benzaldehyde (2) Benzoic acid

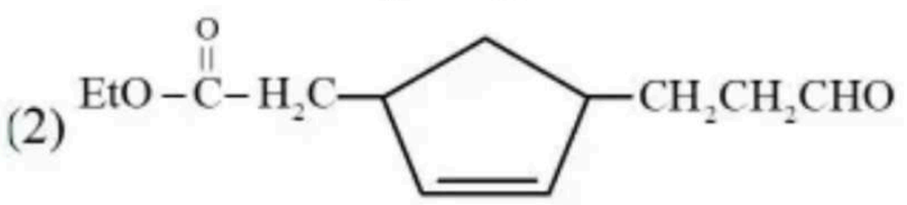
(3) Benzene (4) Toluene

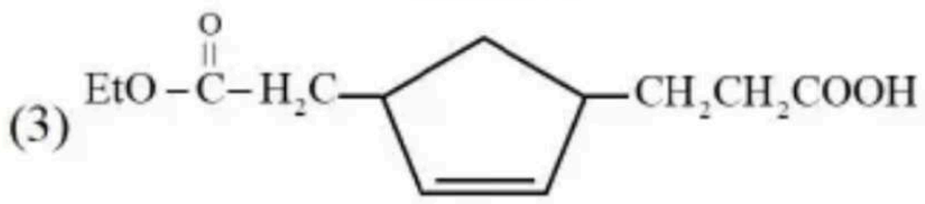
84.

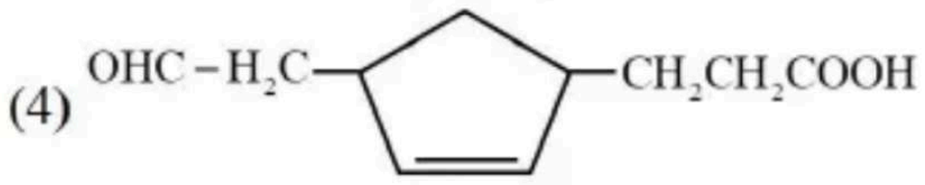


Consider the above reaction and predict the major product.

(1) 

(2) 

(3) 

(4) 

85. 2-Pentanone and 3-pentanone can be distinguished by:

(1) Cannizzaro's reaction

(2) Aldol condensation

(3) Iodoform reaction

(4) Clemmensen's reduction

86. Total number of compounds given below that give positive haloform test are:

CH_3CH_2OH, $CH_3CH_2CH_2OH$, $CH_3CH(OH)CH_3$, HCHO, CH_3CH_2CHO, CH_3CHO, CH_3COCH_3

(1) 4 (2) 5

(3) 3 (4) 7

87. A salt on treatment with dil.H_2SO_4 liberates a colourless gas which turns acidified potassium dichromate solution green. The reaction indicates the presence of

(1) CO_3^{2-} (2) SO_4^{2-}

(3) SO_3^{2-} (4) NO_3^-

88. The correct order of the basic strength of methyl substituted amines in aqueous solution is:

(1) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$

(2) $(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH$

(3) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2$

(4) $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$

89. The reagents that can be used to convert benzenediazonium chloride directly to benzene are _____.
(i) $SnCl_2/HCl$ (ii) CH_3CH_2OH
(iii) $H_3PO_2 + H_2O$ (iv) $LiAlH_4$
(1) (i) and (ii) (2) (ii) and (iii)
(3) (iii) and (iv) (4) (i) and (iii)

90. Glycosidic linkage between C_1 of α-D-glucose and C_2 of β-D-fructose is found in
(1) Maltose (2) Sucrose
(3) Lactose (4) Amylose

BIOLOGY

91. Which of the following is an example of convergent evolution?
(1) Numbat – Anteater
(2) Lemur – Mole
(3) Mole – Bobcat
(4) Spotted cuscus – Numbat

92. Identify the mis-match.
(1) Canidae - Carnivora
(2) Convolvulaceae - Polymoniales
(3) *Panthera* - Felidae
(4) *Musca domestica* – Chordata

93. Match the column I with column II

	Column I		Column II
a.	*Fasciola*	(i)	Metameres
b.	*Laccifer*	(ii)	Flame cells
c.	*Aplysia*	(iii)	Malpighian tubules
d.	*Nereis*	(iv)	Radula

(1) a-(i), b-(ii), c-(iii), d-(iv)
(2) a-(iv), b-(iii), c-(ii), d-(i)
(3) a-(ii), b-(iii), c-(iv) d-(i)
(4) a-(iii), b-(i), c-(ii), d-(iv)

94. Given below are two statements:
Statement-I: In case of *Sycon*, the spongocoel and canals are lined by flagellated cells called choanocytes.
Statement-II: Sponges exhibit external fertilisation and indirect development having a larval stage which is morphologically distinct from the adult.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both statements I and II are correct
(2) Statement I is incorrect, but Statement II is correct
(3) Statement I is correct but Statement II is incorrect
(4) Both Statements I and II are incorrect

95. Type of chlorophyll pigments present in *Euglena* are
(1) a, c (2) a, d
(3) a, b (4) b, c

96. Arrange the following chronologically based on evolution
A) Heterospory
B) Embryo formation
C) Formation of complex tissues
D) Seed formation
(1) B, C, A, D (2) C, A, B, D
(3) B, A, C, D (4) D, C, A, D

97. Match the list-I with list-II
List-I List-II
A) *Neophron* i) Sweat glands
B) *Pristis* ii) Bidder's canal
C) *Macaca* iii) Epidermal scales
D) *Rana* iv) Claspers
(1) A–iii, B–ii, C–iv, D–i
(2) A–iii, B–iv, C–i, D–ii
(3) A–i, B–iv, C–iii, D–ii
(4) A–iii, B–iv, C–ii, D–i

98. *Cycas* corolloid roots contain the following symbiotic microorganism.
(1) *Glomus* (2) *Azatobacter*
(3) *Rhizobium* (4) *Nostoc*

99. Which of the following hormone is not a single aminoacid derivative?
(1) Epinephrine (2) Dopamine
(3) Vasopressin (4) Melatonin

100. The following is not concerned with compound leaf
(1) It may contain a common axis on which leaflets are present
(2) It may contain petiole where the leaflets are attached at the tip
(3) Bud is present in the axil of leaflet
(4) Silk cotton contain palmately compound leaves

101. Identify the incorrect statements from the following

A) The ovary is half inferior in peach

B) In *Cassia*, the corolla shows imbricate aestivation

C) A sterile stamen is called pistillode

D) *Citrus* exhibits polyadelphous stamens

E) In free central placentation the ovary is unilocular

(1) C, E (2) C

(3) B, C, E (4) B, C

102. Identify the correct match.

	List-I		List-II
A.	Asteraceae	I.	Siliqua
B.	Brassicaceae	II.	Caryopsis
C.	Poaceae	III.	Cypsela
D.	Malvaceae	IV.	Epicalyx

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-I, D-II

(2) A-II, B-I, C-III, D-IV

(3) A-I, B-IV, C-II, D-III

(4) A-III, B-I, C-II, D-IV

103. Which of the following are relatively the most recently evolved vertebrates?

(1) Snakes (2) Tuataras

(3) Turtles (4) Dinosaurs

104. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R :

Assertion (A) : Monocot stem can not be differentiated into epidermis, cortex and stele.

Reason (R) : In monocot stem endodermis, pericycle are absent, large number of scattered vascular bundles are present in ground tissue.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is not the correct explanation of A

(3) A is correct but R is not correct

(4) A is not correct but R is correct

105. Read the following

i) It reduces the glucose uptake and utilization by cells

ii) Receptors of this hormone are present in nucleus

iii) It stimulates glycogenesis

iv) It helps in rapid movement of glucose from blood to hepatocytes and adipocytes

Which of the above are true for the hormone secreted by β-cells of Islets of Langerhans?

(1) i and iii (2) i, ii and iv

(3) ii, iii and iv (4) iii and iv

106. Identify the correct statements for the following.

A) The plasmid DNA confers resistance to antibiotics.

B) Pili and fimbriae play a role in motility.

C) Gas vacuoles are inclusion bodies found in blue green, purple and green photosynthetic bacteria.

D) Centrioles are absent in almost all plant cells..

E) In humans, membrane of the erythrocyte has approximately 40% protein and 52% lipids.

(1) Except B (2) A, B, C

(3) Except B, E (4) A, B, E

107. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): Hyperparathyroidism leads to formation of stones in kidneys

Reason (R): PTH stimulates the process of mineralization of bone

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both A and R are correct but R is NOT the correct explanation of A

(2) Both A and R are correct and R is the correct explanation of A

(3) A is not correct but R is correct

(4) A is correct but R is not correct

108. The following is not a function of Golgi complex
(1) Packaging materials
(2) Formation of glycoproteins and glycolipids
(3) Cell plate formation in plant cells
(4) Synthesis of glycogen

109. The following is not a common character between chloroplast and mitochondrion
(1) Semiautonomous cell organelle
(2) double membrane envelope
(3) Presence of ATP synthase
(4) Circular single strand DNA molecule

110. Which of the following accessory reproductive gland's secretion associate with sperms firstly?
(1) Prostate (2) Seminal vesicles
(3) Cowper's glands (4) Bartholin's glands

111. Given below are two statements:
Statement-I: The male sex accessory ducts help in storage, maturation and transportation of sperms.
Statement-II: The ejaculatory ducts loop over the urinary bladder in males
In the light of the above statements, choose the correct answer from the options given below:
(1) Both statements I and II are correct
(2) Statement I is incorrect, but Statement II is correct
(3) Statement I is correct but Statement II is incorrect
(4) Both Statements I and II are incorrect

112. Identify the correct match.

	List-I		List-II
A.	Palmitic acid	I.	20 carbons
B.	Glycine	II.	Nucleoside
C.	Arachidonic acid	III.	16 carbons
D.	Uridine	IV.	Nucleotide
		V.	2 carbons

Choose the correct answer from the options given below :
(1) A-III, B-IV, C-I, D-II
(2) A-II, B-I, C-III, D-IV
(3) A-I, B-IV, C-II, D-III
(4) A-III, B-V, C-I, D-II

113. Which one of the following statement is wrong w.r.t female human
(1) The lactiferous ducts of each mammary lobe join to form mammary ampulla.
(2) Several mammary ducts join to form a mammary ampulla.
(3) Ovary is connected to pelvic wall and uterus by ligaments
(4) Isthmus is the last part of oviduct

114. Given below are two statements :
Statement-I : GLUT-4 is a lipid enables glucose transport into cells.
Statement-II : Proteins are heteropolymers madeup of 20 types of amino acids.
In the light of the above statements, choose the correct answer from the options given below :
(1) Both Statement I and Statement II are true
(2) Statement I is false but Statement II is true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false

115. Leydig cells are regulated by the pituitary hormone known as
(1) FSH (2) GnRH
(3) Testosterone (4) LH (ICSH)

116. *Escherichia coli* with completely heavy DNA ^{15}N was allowed to replicate in a medium with $^{14}NH_4Cl$ for two generations. The percentage of bacteria with ^{15}N in its DNA is
(1) 25 (2) 12.5
(3) 50 (4) 6.25

117. Which of these is not an autoimmune disease?
(1) Myasthenia gravis
(2) SCID
(3) Addison's disease
(4) Rheumatoid arthritis

118. Identify the incorrect regarding enzymes.
(1) Enzymes bring down activation energy barrier making the transition of 'S' to 'P' more easily.
(2) Km value is inversely proportional to affinity of enzyme with its substrate molecules.
(3) Competitive inhibitors are often used in the control of bacterial pathogens.
(4) Low temperature denatures the enzyme.

119. Which of the following statements are correct about cancer cells?
(i) Do not show the property of contact inhibition
(ii) Cell growth and differentiation is highly controlled
(iii) Actively divide and compete with normal cells for vital nutrients
(iv) Remain confined to their original location
(v) Cells sloughed from the tumor reach distant sites and start forming new tumor there
(1) Only (i) and (v) (2) Only (ii), (iv), (v)
(3) Only (i), (iii), (v) (4) Only (i) and (v)

120. Which of the following is incorrect about cannabinoids?
(1) Affects cardiovascular system of body
(2) Flower tops, leaves and resin of *Cannabis* plant are used in various combinations to produce marijuana, hashish, charas and ganja
(3) Interferes with the transport of neurotransmitter dopamine
(4) Interacts with cannabinoid receptors present in brain

121. Given below are two statements :
Statement-I : M phase is the most dramatic period of the cell cycle, involving a major reorganisation of virtually all components of the cell.
Statement-II : In 'S' phase the duplicated centrioles begin to move towards opposite poles of the cell.
In the light of the above statements, choose the correct answer from the options given below :
(1) Both Statement I and Statement II are true
(2) Statement I is false but Statement II is true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false

122. Which of the following is not the effect of tobacco smoking?
(1) Increased secretion of catecholamine hormones
(2) Decreased blood pressure
(3) Causes oxygen deficiency in the body
(4) Causes coronary heart diseases

123. Arrange the following events in sequence which occur during meiosis.
A) Segregation of chromosomes
B) Crossing over
C) Terminalization
D) Synapsis
E) Division of centromere
(1) D, B, C, A, E (2) C, A, B, D, E
(3) A, B, C, D, E (4) D, B, A C, E

124. Choose the correct statement regarding skeletal muscle
(1) Myofibrils of a muscle cell are held together by a common collagenous connective tissue layer called fascia
(2) Nervous system does not control their activities directly
(3) A characteristic feature of the muscle fibre is the presence of parallelly arranged myofilaments in the sarcoplasm
(4) The isotropic bands contain myosin whereas anisotropic bands contain actin proteins only

125. Spindle fibers are attached to
(1) Kinetochores
(2) Telomeres
(3) Secondary constriction
(4) Satellite

126. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion(A): The muscles located in the walls of visceral organs like alimentary canal are smooth in appearance
Reason(R): Activities of smooth muscles are not under the voluntary control of the nervous system
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both A and R are correct and R is the correct explanation of A
(2) Both A and R are correct but R is NOT the correct explanation of A
(3) A is not correct but R is correct
(4) A is correct but R is not correct

127. If karyokinesis is not followed by cytokinesis leading to the formation of
(1) Polyploid (2) Monosomics
(3) Nullisomics (4) Syncytium

128. Identify the correct statement from the following regarding non-cyclic electron transport.
A) The movement of electrons is downhill
B) Ultimate donor of electrons is water molecule.
C) Final electron acceptor is $NADP^+$.
D) Electrons moves along the gradient of energy.
E) Electrons are transferred to another acceptor molecule that has a greater redox potential
(1) A, B, C only (2) B, D, E only
(3) A, C, D only (4) All the above

129. Match the following column I with column II

Column-I	Column-II (Number of bones)
a) Cranium	i) 22
b) Vertebral column	ii) 4
c) Axial skeleton	iii) 80
d) Pectoral girdle	iv) 26
	v) 8

(1) a-iv, b-iii, c-ii, d-i (2) a-iii, b-iv, c-i, d-ii
(3) a-i, b-iv, c-iii, d-ii (4) a-v, b-iv, c-iii, d-ii

130. Identify the correct statements and choose the appropriate option accordingly
(i) Reabsorption of water occurs passively in the initial segments of nephron
(ii) Nitrogenous wastes are reabsorbed by active transport
(iii) Conditional reabsorption of Na and water takes place in DCT.
(iv) Reabsorption of Ca^{2+} occurs at DCT
(v) DCT is capable of reabsorption of HCO_3^- and selective secretion of H^+, K^+ and NH_3 to maintain pH and ionic balance in blood.
(1) Only (i) and (ii) (2) Only (iii) and (iv)
(3) Only (iv) and (v) (4) (i), (iii), (iv), (v)

131. How many turns of Calvin cycle are required to produce one net glucose molecule ?
(1) 6 (2) 1
(3) 12 (4) 3

132. During C_4 cycle the secondary carboxylation occur in
(1) Cytoplasm of mesophyll cell
(2) Chloroplast of mesophyll cell
(3) Cytoplasm of Bundle sheath cell
(4) Chloroplast of Bundle sheath cell

133. Study the following statements and choose the correct option.
a) During power stroke overlapping region of myofilaments decreases
b) Length of 'I' band decreases during power stroke
c) Subunit of troponin masks the active binding sites for myosin on the thin filaments in resting state.
d) Binding of new ATP to the myosine head causes breakdown of cross bridge during mechanism of muscle contraction
(1) All are true except d
(2) All are true except b
(3) All are true except c
(4) All are true except a

134. Match the column I and II, and choose the correct combination from the options given below

Column -I	Column -II
a) Sterols and wax	i) Kidneys
b) Water and CO_2	ii) Liver
c) Urea and lactic acid	iii) Sweat glands
d) Degraded steroid hormones	iv) Lungs
e) Creatinine	v) Sebaceous glands

(1) a–iv, b–v, c–i, d–ii, e–iii
(2) a–v, b–iv, c–iii, d–ii, e–i
(3) a–i, b–ii, c–iii, d–iv, e–v
(4) a–v, b–ii, c–iii, d–iv, e–i

135. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R :
Assertion (A) : NADH is oxidised to NAD^+ rather slowly in fermentation, however the reaction is very vigorous in case of aerobic respiration.
Reason (R) : Glycerol would enter the respiratory pathway after being converted to PGAL.
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both A and R are correct and R is the correct explanation of A
(2) Both A and R are correct but R is not the correct explanation of A
(3) A is correct but R is not correct
(4) A is not correct but R is correct

136. MacArthur showed that five closely related species of warblers living on the same tree were able to avoid competition and co-exist due to?
(1) Physiological differences
(2) Behavioural differences in their foraging activities
(3) Anatomical differences in immune system
(4) Biochemical differences

137. In mitochondria the copper centers are present in
(1) Complex IV (2) Complex II
(3) Complex I (4) Complex III

138. Which of the following is not correctly matched?
(1) Hyaline cartilage – Tracheal rings
(2) Collagen – Intervertebral disc
(3) Elastic cartilage – Epiglottis
(4) Elastic cartilage – Costal cartilage of rib

139. Identify the correct match

	List-I		List-II
A.	Ethylene	I.	Promotes female flowers in cucumbers
B.	Auxins	II.	Overcome apical dominance
C.	ABA	III.	Cause apical dominance
D.	Cytokinins	IV.	Stress hormone

Choose the correct answer from the options given below :
(1) A-III, B-IV, C-I, D-II
(2) A-II, B-I, C-III, D-IV
(3) A-I, B-III, C-IV, D-II
(4) A-III, B-II, C-I, D-IV

140. Choose the incorrect match w.r.t. cockroach.
(1) Gonapophyses – Chitinous and asymmetrical
(2) Each ovary – Eight ovarioles
(3) Collaterial glands – Form spermatophores
(4) Development – Paurometabolous

141. The following belongs to terpenes
(1) ^6N Furfurylamino purine
(2) Gibberellic acid
(3) Ethylene
(4) Indole 3 acetic acid

142. Which of the following is incorrect statement w.r.t. frog?
(1) Sense organs of touch, taste and smell are cellular aggregations around nerve endings
(2) External ear is absent and only tympanum can be seen externally and it helps in only balancing
(3) Midbrain is characterized by corpora bigemina
(4) A pair of simple eyes are present

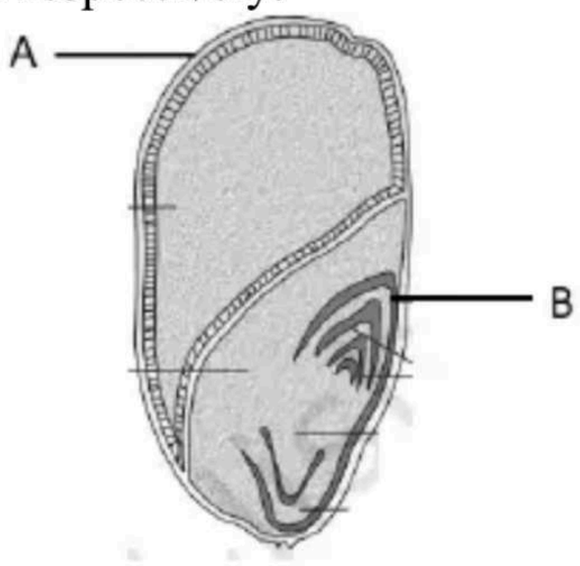
143. The initial length of stem is 20 centimeters. After specific time the length is 30 centimeters. What is the RGR ?
(1) 10% (2) 20%
(3) 25% (4) 50%

144. Match the following:

List – I	List – II
a) Tendons and ligaments	i) Areolar connective tissue
b) Dermis of skin	ii) Dense irregular connective tissue
c) Subcutaneous layer	iii) Areolar and adipose connective tissue
d) Supporting framework for epithelium	iv) Dense regular connective tissue

(1) a-ii, b-iv, c-i, d-iii (2) a-iv, b-ii, c-i, d-iii
(3) a-i, b-ii, c-iv, d-iii (4) a-iv, b-ii, c-iii, d-i

145. Study the following diagram and identify A and B respectively.



A and B are

(1) Endosperm, coleoptile

(2) Scutellum, Radicle

(3) Pericarp, coleoptile

(4) Coleorhiza, plumule

146. Autogamy can not be seen in

(1) Bisexual flowers

(2) Monoecious plants only

(3) Dioeecious plants only

(4) Both monoecious and dioecious plants

147. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): The walls of ventricles of heart are much thicker than that of the atria.

Reason (R): The atrium and ventricle of the same side are separated by a thick fibrous tissue called the atrio – ventricular septum

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is NOT the correct explanation of A

(3) A is not correct but R is correct

(4) A is correct but R is not correct

148. An individual having AB blood group type

(1) can receive only AB group blood

(2) can donate to all types of blood groups of ABO blood group system

(3) can receive blood from all four types of blood groups of ABO blood group system

(4) can receive blood from A/B/AB blood group person, but not from "O" group person

149. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R :

Assertion (A) : ABO blood grouping in human show, dominance, co-dominance and multiple alleles.

Reason (R) : ABO blood grouping is controlled by 'I' gene having three alleles and $I^A = I^B > i$.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is not the correct explanation of A

(3) A is correct but R is not correct

(4) A is not correct but R is correct

150. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): The movement of air into and out of the lungs is carried out by creating a pressure gradient between the lungs and the atmosphere.

Reason (R): Inspiration can occur if the intra-pulmonary pressure is greater than the atmospheric pressure.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is NOT the correct explanation of A

(3) A is not correct but R is correct

(4) A is correct but R is not correct

151. Identify incorrect regarding linkage

(1) The distance between two genes on a chromosome is directly proportional to the recombination frequency between them

(2) Genetic maps are extensively used as a starting point in the sequencing of whole genomes as was done in case of human genome sequencing project

(3) All the genes present on one chromosome form one linkage group

(4) Morgan carried out several monohybrid crosses in *Drosophila* to study genes that are sex linked

152. Identify the mismatched combination

(1) Immune response – B cells

(2) Blood coagulation – Platelets

(3) Polymorpho-nuclear leucocytes – Monocytes

(4) Inflammatory reactions - Basophils

153. If skin colour in humans is controlled by three genes pairs, then what is the number of parental genotypes in F_2 generation ?

(1) 62/64 (2) 2/64

(3) 14/64 (4) 1/14

154. What will be the total lung capacity of a person with vital capacity of 4000 mL, and a residual volume of 1000mL?

(1) 4500 mL (2) 5000 mL

(3) 5500 mL (4) 6000 mL

155. Given below are two statements :

Statement-I : In DNA molecules the nitrogenous bases that are linked to sugar moiety project inside from the backbone.

Statement-II : During DNA replication process the incoming nucleotide forms a phosphoester bond with 3 OH group of sugar molecule.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

156. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): The number of action potentials that could be generated in a minute vary at different parts of the nodal system

Reason (R): Cardiac muscles are striated and are highly resistant to fatigue

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is NOT the correct explanation of A

(3) A is not correct but R is correct

(4) A is correct but R is not correct

157. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R :

Assertion (A) : Thymine is more stable when compared to uracil.

Reason (R) : Thymine has a methyl group at 5′ carbon atom which increases chemical stability of base.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is not the correct explanation of A

(3) A is correct but R is not correct

(4) A is not correct but R is correct

158. If a clourblind boy has normal brother and clourblind sister, then their parents are

A) Normal father and clourblind mother

B) Both mother and father are normal

C) Father is clourblind and mother is carrier

D) Both parents are clourblind

E) Father is colourblind and mother is homozygous normal

Choose the most appropriate answer from the options given below.

(1) Only A and B are correct

(2) Only B and C are correct

(3) Only D and E are correct

(4) Only C is correct

159. Identify the correct statement.
(1) The semi-conservative mode of DNA replication was first shown in eukaryotes
(2) In Meselson and Stahl experiment N^{14} and N^{15} isotope used are radioactive
(3) In Meselson and Stahl experiment after 40 min. both heavy and hybrid DNA molecules are formed
(4) Okazaki fragments are joined by DNA ligase
(5) During transcription DNA double helix is opened by RNA polymerase
How many of the above statements are correct?
(1) 5 (2) 4
(3) 2 (4) 3

160. Find out the incorrectly matched genetic disorder and its associated chromosome
(1) Myotonic dystrophy – Chromosome 19
(2) Alkaptonuria – Chromosome 3
(3) Tay Sach's disease – Chromosome 15
(4) Cystic fibrosis – Chromosome 9

161. The following amino acids do not show degeneracy
(1) Cysteine, Glycine
(2) Valine, Threonine
(3) Proline, Serine
(4) Methionine, Tryptophan

162. Given below are two statements:
Statement-I: When both parents are heterozygous regarding thalassemia, their children may be affected by the disorder
Statement - II: Thalasemia is an autosomal dominant disorder.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both statements I and II are correct
(2) Statement I is incorrect, but Statement II is correct
(3) Statement I is correct but Statement II is incorrect
(4) Both Statements I and II are incorrect

163. Enzyme that transcribes precursor of mRNA (Primary transcript) in eukaryotes
(1) RNA polymerase IV
(2) RNA polymerase II
(3) RNA polymerase I
(4) RNA polymerase III

164. During termination of polypeptide chain release factor binds to
(1) AUG (2) GGG
(3) UUU (4) UAA

165. Which one of the following statements about certain given animals is correct?
(1) Circulatory system is of open type in *Saccoglossus*
(2) Body of *Limulus* is covered by calcareous spines
(3) Body of tape worm is circular in cross section
(4) Molluscs are mostly viviparous

166. Match List-I with List-II :

	List-I		List-II
A.	rop	I.	Modified *E.coli* vector
B.	Tet^R	II.	Pvu I
C.	Amp^R	III.	BamH I
D.	pBR322	IV.	Pvu II

Choose the correct answer from the options given below :
(1) A-III, B-IV, C-I, D-II
(2) A-II, B-I, C-III, D-IV
(3) A-IV, B-III, C-II, D-I
(4) A-III, B-II, C-I, D-IV

167. If the frequency of recessive allele in the Hardy-Weinberg population is 40%. Find the number of dominant individual out of a population of 5000 individuals-
(1) 1800 (2) 4200
(3) 3500 (4) 2400

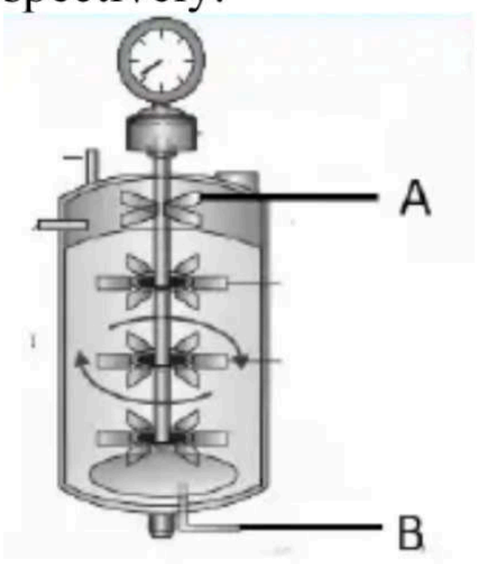
168. Given below are two statements :
Statement-I : The separated bands of DNA are cut out from agarose gel and extracted from the gel piece. This step is known as elution.
Statement-II : DNA that separates out after adding chilled ethanol is removed by spooling.
In the light of the above statements, choose the correct answer from the options given below :
(1) Both Statement I and Statement II are true
(2) Statement I is false but Statement II is true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false

169. According to the theory of spontaneous generation, life came out
(1) In the form of spores from different planets
(2) Of decaying and rotting matter like straw, mud etc.
(3) From pre-existing non-living inorganic molecules (eg. RNA, Protein etc.)
(4) From pre-existing life

170. During PCR primers are
(1) single stranded oligonucleotides with ribose sugar
(2) single stranded oligonucleotides with deoxyribose sugar
(3) double stranded oligonucleotides with deoxyribose sugar
(4) double stranded oligonucleotides with ribose sugar

171. Which one of the following birth-control methods has maximum chance of failure of contraception?
(1) Vasectomy
(2) Contraceptive pills
(3) Lactational amenorrhea
(4) LNG – 20

172. Study the following diagram and identify A and B respectively.



(1) Motor, Sterile Air
(2) Sterile air, Foam braker
(3) Foam braker, Sterile Air
(4) Nutrient broth, Sterile Air

173. Which one of the following cannot be cured by antibiotics?
(1) Syphilis (2) Gonorrhoea
(3) Chlamydiasis (4) Genital herpes

174. In insertional inactivation method the transformant recombinant colonies of bacterial cells appear in
(1) blue colour (2) white colour
(3) orange colour (4) red colour

175. Study the following
a) Presence of nerve ring b) Antennae
c) Legs d) Segmental ganglia
e) Wings
Which of the above are related to prothorax of cockroach?
(1) c, d (2) b, c, d
(3) b, c, e (4) a, b, d

176. Select the incorrect match.
(1) Acetic acid – *Acetobacter aceti*
(2) Citric acid – *Aspergillus niger*
(3) Streptokinase - *Streptococcus*
(4) Statins – *Mucor*

177. cry II Ab and cry I Ab produced toxins that control ____ and _____ respectively
(1) corn borer, cotton bollworm
(2) beetles, borer
(3) flies, mosquitoes
(4) cotton bollworm, corn borer

178. Select the correct sequence of the following places based on the number of bird species-
[A - India, B - Greenland, C - Colombia, D - New York]
(1) D > C > B > A (2) A > C > D > B
(3) C > A > D > B (4) C > D > B > A

179. Match the animals given in column I with their extinction location in column II.

	Column-I		Column-II
(a)	Dodo	(i)	Africa
(b)	Quagga	(ii)	Russia
(c)	Thylacine	(iii)	Mauritius
(d)	Stellar's sea cow	(iv)	Australia

(1) (a)-(i), (b)-(iii), (c)–(ii), (d)-(iv)
(2) (a)-(iv), (b)-(iii), (c)-(i), (d)–(ii)
(3) (a)-(iii), (b)-(i), (c)–(ii), (d)-(iv)
(4) (a)-(iii), (b)-(i), (c)-(iv), (d)–(ii)

180. Statement-I: In a terrestrial ecosystem, detritus food chain is the major conduit for energy flow.
Statement-II: DFC begins with dead organic matter
In the light of the above statements, choose the correct answer from the options given below:
(1) Both statements I and II are correct
(2) Statement I is incorrect, but Statement II is correct
(3) Statement I is correct but Statement II is incorrect
(4) Both Statements I and II are incorrect